

1. **Hybridoma cells produce the following on an industrial scale?**
 - a. Cancer cells
 - b. Antibodies
 - c. Antigens
 - d. Cell lines
2. **Monoclonal antibodies (Mabs) can be used to purify the following molecules?**
 - a. RNA
 - b. Lipids
 - c. DNA
 - d. Proteins
3. **In ELISA, inefficient washing of the wells between incubation steps is a common cause of?**
 - a. Non-specific variability
 - b. Not valid unnecessary step
 - c. Specificity
 - d. False negative results
4. **Based on morphology, the lymphoblast-like cells have?**
 - a. Spherical in shape suspended in culture
 - b. Usually have irregular shapes
 - c. Polygonal in shape with regular dimensions
 - d. Bipolar elongated shapes
5. **The following describe the continuous cell lines EXCEPT:**
 - a. Slow growth
 - b. Have aneuploidy chromosome number
 - c. Are anchorage independent
 - d. Short generation time (12–24 h)
6. **The following describe suspension cells EXCEPT?**
 - a. Comprise hematopoietic cell lines
 - b. Present the majority of cells derived from vertebrates
 - c. Free floating in culture

- d. Have a short lag period
- 7. **Normal cells usually divide only a limited number of times before losing their ability to proliferate, which is a genetically determined event known as senescence. These cell lines are called?**
 - a. Continuous
 - b. Finite
 - c. Cancerous
 - d. Transformed
- 8. **Culture media is usually stored at?**
 - a. Liquid nitrogen
 - b. -20°C
 - c. -80°C
 - d. 4°C
- 9. **For successful monoclonal antibody production?**
 - a. Male mice must be used in immunization
 - b. Males produce more antibodies than females
 - c. Female mice must be used in immunization
 - d. Male or female mice can be used in immunization
- 10. **In order to avoid tolerance, the average dose for mice immunization is?**
 - a. 1-10 microgram
 - b. 100-200 microgram
 - c. 10-100 milligram
 - d. 10-100 microgram

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
B	D	A	A	A	B	B	C	D	D

Questions (Continued):

11. The following characterize Mabs in comparison to polyclonal antibodies:

- a. Prone to batch-to-batch variability
- b. Preferred for detection of denatured proteins
- c. Expensive to produce
- d. Recognize multiple epitopes

12. The best method of RPMI-media sterilization is by using?

- a. Dry air in oven at 160°C
- b. Filtration using 0.45 µm followed by 0.2 µm filters
- c. Boiling for 20 min
- d. Addition of 70% alcohol

13. Hybridoma cells are produced from the fusion of the following cells?

- a. A specific B cell and a nonspecific myeloma cell
- b. A nonspecific B cell and a nonspecific myeloma cell
- c. A nonspecific B cell and a specific myeloma cell
- d. A specific B cell and a specific myeloma cell

14. The following sentence is correct about the proper concentration of fetal calf serum (FCS) during the process of hybridoma production?

- a. 0% FCS in washing media
- b. 10% FCS in the fusion between myeloma and B cells
- c. 20% FCS in culturing of hybridoma cells
- d. 5% FCS in hybridoma storage in liquid nitrogen

15. The following makes a limitation to using monoclonal antibodies (Mabs) in therapeutic purposes?

- a. Development of anti-mouse antibodies
- b. All choices are correct
- c. Monoclonal antibodies may not be attached to some cancer cells
- d. The free antigens present in the circulation may bind to Mabs and prevent them from their action

16. Among the weekly tasks in a tissue culture lab?

- a. Washing the prefilter

- b. Dismantle the inside of the incubator and wash it with alcohol
- c. Mopping the floor with detergent
- d. Wipe all general surfaces with 70% alcohol

17. ELISA stands for?

- a. Enzyme-linked inhibition assay
- b. Enzyme-linked immunosorbent assay
- c. Enzyme light immune sandwich assay
- d. Enzyme-like inhibition substrate assay

18. To test for the stability of the complete Freund adjuvant, a small drop of the emulsion is applied on the surface of?

- a. Water
- b. Methanol
- c. Chloroform
- d. Alcohol

19. The following is correct to ensure aseptic conditions while working in a laminar flow hood?

- a. Crossing your arms over the opened tissue culture flasks
- b. Putting the pipette aid in the center of the hood while the media flask is on the right side
- c. Interrupting the airflow inside the hood
- d. Wearing sterile gloves is recommended as dead skin cells of your hand can be a potential source of contamination

Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19
C	B	A	C	B	D	B	A	D

20. The following characterize Mabs in comparison to polyclonal antibodies:

- a. Prone to batch-to-batch variability
- b. Expensive to produce
- c. Recognize multiple epitopes
- d. Preferred for detection of denatured proteins

21. In ELISA, to test for nonspecific label binding (Antibody-enzyme conjugate), the following control is used?

- a. Irrelevant antibody
- b. Alternative antigen
- c. Different enzyme conjugate
- d. Known positive antibody
- e. Buffer

22. The following is correct to ensure aseptic conditions while working in a laminar flow hood?

- a. Interrupting the airflow inside the hood
- b. Crossing your arms over the opened tissue culture flasks
- c. Wearing sterile gloves is recommended as dead skin cells of your hand can be a potential source of contamination
- d. Putting the pipette aid in the center of the hood while the media flask is on the right side

23. For successful monoclonal antibody production?

- a. Males produce more antibodies than females
- b. Male mice must be used in immunization
- c. Male or female mice can be used in immunization
- d. Female mice must be used in immunization

24. Normal cells usually divide only a limited number of times before losing their ability to proliferate, which is a genetically determined event known as senescence. These cell lines are called?

- a. Transformed
- b. Finite
- c. Continuous
- d. Cancerous

25. The antigen preparation for injection for the successful production of monoclonal antibodies?

- a. Must be pure
- b. Must have high affinity and high concentration
- c. Can be pure or not pure
- d. Must be impure

26. The following describe suspension cells EXCEPT?

- a. Have a short lag period
- b. Present the majority of cells derived from vertebrates
- c. Free floating in culture
- d. Comprise hematopoietic cell lines

27. ELISA is?

- a. Less sensitive than double diffusion
- b. Less sensitive than single immunodiffusion
- c. More sensitive than double diffusion
- d. Less sensitive than immunoelectrophoresis

28. Culture media is usually stored at?

- a. 4°C
- b. Liquid nitrogen
- c. -20°C
- d. -80°C

29. The following describe the adherent cells in tissue culture EXCEPT?

- a. While passaged, a detaching agent (trypsin/EDTA) is used to cover the monolayer
- b. Cells which are anchorage dependent and propagate as monolayer
- c. Include fibroblasts and epithelial cells

Q20	Q21	Q22	Q23	Q24	Q25	Q26	Q27	Q28	Q29
B	A	C	C	B	C	B	C	D	A

30. Class I laminar flow?

- a. Offer significant levels of protection to laboratory personnel
- b. Suitable for hybridoma culture preparation of monoclonal antibodies
- c. Hoods provide an aseptic environment necessary for cell culture experiments
- d. Provide cultures protection from contamination or cells of human origin

31. The following characterize polyclonal antibodies in comparison to Mabs?

- a. Higher affinity
- b. Less background relative to Mabs
- c. Bind only one epitope
- d. Consistency between different lots

32. The following describe tissue culture contamination by viruses?

- a. Is detected by testing the cultures using fluorescent Hoechst staining
- b. Are the most commonly encountered biological contaminants in cell culture
- c. Viral infection of cell cultures can be detected by ELISA assays, or PCR
- d. The best and fast diagnostic method is by microscopy

Q30	Q31	Q32
A	A	c

33. The requirement for continued storage of cells at all stages of Mab production is essential because?

- a. There is always a risk of overgrowth by Ig-nonproducing hybridomas
- b. Throughout cloning, there is a high probability of chromosome loss and consequent loss of Ig secretion
- c. All choices are correct
- d. The risk of infection

34. **To test whether the separated hybridoma clones are identical or totally different, the following test could be performed?**
- a. Specific immunoelectrophoresis
 - b. Competitive inhibition radioimmunoassay
 - c. Specific agglutination test
 - d. Specific double immunodiffusion
35. **For separation of specific from nonspecific human B cells, the following positive selection method is effective?**
- a. Ficoll-hypaque separation method
 - b. Ammonium sulfate precipitation method
 - c. ELISA using specific Abs
 - d. Magnetic beads coated with Abs against CD19 is used
36. **The following are a list of essential components of cell culture media. Match them to the requirements for effective cell culture which they fulfil?**
- a. Bicarbonate
 - b. Mercaptoethanol
 - c. Phenol red
 - d. Inorganic salts
 - e. Glutamine
37. **Human myelomas are difficult to establish in tissue culture because all of the following EXCEPT?**
- a. Are large size
 - b. Have abundant RER and mitochondria with high Ig production
 - c. Doubling time in culture is about 70 hours
 - d. Do not carry EBV
38. **The following describe adherent cell line?**
- a. Free floating in culture
 - b. Comprise hematopoietic cell lines
 - c. Deposit at the walls of the culture
 - d. Present the majority of cells derived from vertebrates

39. Class I laminar flow?

- a. Is not suitable for hybridoma culture preparation of monoclonal antibodies
- b. Hoods provide an aseptic environment necessary for cell culture experiments
- c. Provide the highest attainable level of protection is required for work involving known human pathogens
- d. Provide cultures protection from contamination or cells of human origin

40. Among the weekly tasks in tissue culture lab?

- a. Mopping the floor with detergent
- b. Wipe all general surfaces with 70% alcohol
- c. Pouring disinfectant in the sink
- d. Washing the prefilter

41. Hybridoma cells have an application to produce?

- a. Antibodies
- b. Antigens
- c. Cancer cells
- d. Cell lines

42. Based on morphology, the lymphoblast-like cells have?

- a. Usually have irregular shapes
- b. Polygonal in shape with regular dimensions
- c. Spherical in shape suspended in culture
- d. Bipolar elongated shapes

Q33	Q34	Q35	Q36	Q37	Q38	Q39	Q40	Q41	Q42
C	B	D	E	D	C	B	B	A	C

43. The following describe adherent cell line?

- a. Free floating in culture
- b. Comprise hematopoietic cell lines
- c. Deposit at the walls of the culture
- d. Present the majority of cells derived from vertebrates

44. Class I laminar flow?

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- b. Hoods provide an aseptic environment necessary for cell culture experiments
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45. Among the weekly tasks in tissue culture lab?

- a. Mopping the floor with detergent
- b. Wipe all general surfaces with 70% alcohol
- c. Pouring disinfectant in the sink
- d. Washing the prefilter

46. Hybridoma cells have an application to produce?

- a. Antibodies

Q43	Q44	Q45	Q46
C	B	B	A

47. The recommended adjuvant for in vitro immunization is?

- a. Complete Freund adjuvant
- b. Incomplete Freund adjuvant
- c. Muranyl dipeptide
- d. CPG

48. The following describe a suspension cells EXCEPT?

- a. Have a short lag period

- b. Free floating in culture
- c. Present the majority of cells derived from vertebrates
- d. Comprise hematopoietic cell lines

49. The best method of RPMI-media sterilization by using?

- a. Filtration using 0.45 µm followed by 0.2 µm filters
- b. Addition of 70% alcohol
- c. Boiling for 20 min
- d. Dry air in oven at 160°C

50. The following describe ELISA EXCEPT?

- a. The mostly used buffer is carbonate
- b. It is a specific test
- c. The preferred enzyme in antibody conjugate is horseradish collagenase
- d. The best way of reducing the nonspecific binding in ELISA, by incubating the coated plate with bovine serum albumin BSA
- e. The antigen can be attached to the surface of ELISA plates

51. The following characterize Mabs in comparison to polyclonal antibodies?

- a. Prone to batch-to-batch variability
- b. Recognize multiple epitopes
- c. Preferred for detection of denatured proteins
- d. Expensive to produce

Q47	Q48	Q49	Q50	Q51
A	C	A	C	B